ENTERPRISE ARCHITECTURE BLUEPRINT

AI-DRIVEN ENTERPRISE CPQ

Google Cloud reference architecture for value-share pricing and dynamic margin governance

Turn AI productivity into measurable client value - without losing commercial control.



REFERENCE ARCHITECTURE / DESIGN BLUEPRINT

AUDIENCE CTO • CFO • LEAD ENTERPRISE ARCHITECTS

OCTOBER 2026

EXECUTIVE THESIS

The productivity paradox is a commercial design problem

CPQ should convert faster delivery into protected margin and provable client value.

01 THE PRESSURE

GenAI can shorten delivery by 30%-50%. Under legacy Time & Materials contracts, fewer billable hours can erode revenue.

02 THE COMMERCIAL SHIFT

Decouple revenue from effort through outcome-based value-share, software-asset reuse, and automated deal-desk margin floors.

03 THE PLATFORM RESPONSE

Use deterministic pricing rules, AI-assisted deal intelligence, and auditable routing on a globally scalable cloud platform.

TARGET OPERATING ENVELOPE

$84.5B	16.4%-16.6%	< 200 ms	< 2 sec
annual deal flow in scope	target margin floor	interactive p99 target	Tier 1 clearance target

Executive premise **AI-led delivery gains should fund client value and reusable assets - not undermine margin discipline.**

Figures are planning assumptions from the supplied blueprint; validate against current commercial and performance data.

PLATFORM DESIGN

A cloud-native decision plane, with explicit operational guardrails

Keep quote-time decisions deterministic; use AI for inference and recommendations, not unbounded approval authority.

Runtime

SERVERLESS

Cloud Run v2 for API, pricing rules, margin calculation, and Monte Carlo simulation.

Global ingress

EDGE SECURITY

Global External Application Load Balancer, Cloud Armor, IAP, and Workspace OAuth.

Multi-region

RESILIENCE

us-east4 + europe-west3 behind the global load balancer.

Integration

DUAL API

REST/JSON for Salesforce CPQ; gRPC for service-to-service calls.

Warm tiering

LATENCY + COST

2 minimum instances during 07:00-20:00 local; scale to zero off-hours.

Concurrency

PERFORMANCE

Configured concurrency: 80. Validate resource sizing and p99 under realistic peaks.

DESIGN RULE **Interactive quote path stays small and predictable; data enrichment, retraining, and embedding workloads run asynchronously or off-peak.**

COMPUTE CHOICE

Select Cloud Run v2 for the interactive CPQ path

GKE Autopilot remains a viable alternative for workloads that require Kubernetes-level control or batch-specific patterns.

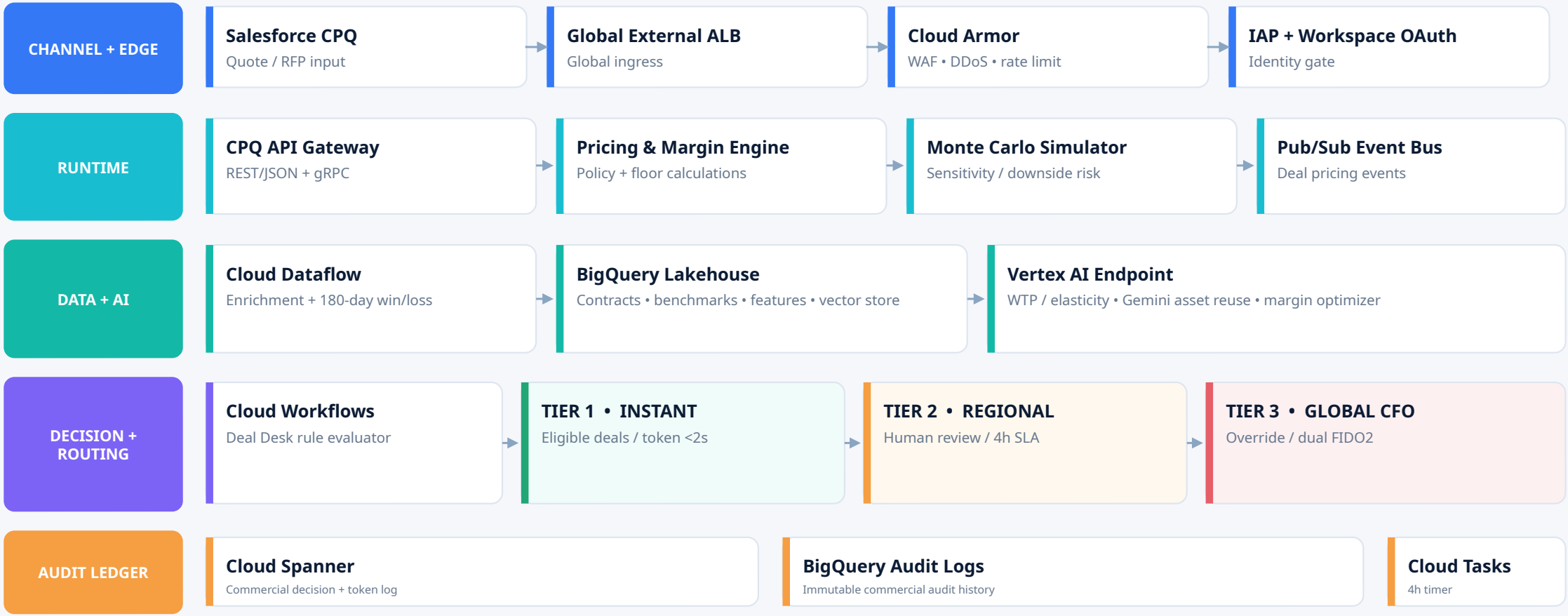
DIMENSION	CLOUD RUN V2 • SELECTED	GKE AUTOPILOT • ALTERNATIVE	ARCHITECTURAL READ-THROUGH
Operations	No node pools, OS patching, or control-plane version management.	Still maintain Kubernetes manifests, CRDs, and disruption budgets.	Favor Cloud Run for deal-desk focus.
Elasticity	Scale to zero off-hours; scale out with quote demand.	Autoscaling has lag and does not remove all baseline cluster costs.	Align capacity to sales activity.
Latency	Warm instances target sub-200 ms interactive requests.	Pod scheduling can take longer; warm overprovisioning may be needed.	Benchmark with production-like spikes.
Cost model	Request-based CPU/RAM billing; idle cost can be minimized.	Pod allocations plus cluster management overhead.	**Source estimates 42% lower TCO for this workload.**
Security	Native IAM and per-service service accounts.	Workload Identity maps Kubernetes SAs to Google SAs.	Both can be secured; Cloud Run is simpler here.

The 42% TCO figure and latency figures are source-stated estimates/targets, not independently verified benchmarks.

REFERENCE ARCHITECTURE

One governed flow from Salesforce quote to auditable commercial decision

Core services use Cloud Run; AI and analytics enrich decisions; Workflows enforces the approval policy.



Governance controls span the quote path; Cloud Tasks starts/escalates the human-review SLA.

DATA + AI

AI informs the price; a constrained solver protects the floor

The source architecture combines client willingness-to-pay, reusable assets, and a deterministic operating-margin constraint.

1 INGEST

RFP scope • industry • spend • timeline • competitor signals
Pub/Sub + Dataflow

→

2 FEATURE

10+ years of outcomes; 180-day win/loss windows; financial and sector context
BigQuery

→

3 INFER

WTP / price elasticity; asset reuse extraction; margin-floor optimization
Vertex AI + Gemini

→

4 GOVERN

Return price sensitivity and reuse factor; apply hard rules before quote emission
Workflows

ASSET REUSE COST ADJUSTMENT

Adjusted COSP = Baseline COSP × (1 - 0.65α)

Example from source: α = 0.342 → modeled delivery-cost reduction ≈ 22.2%.

OPERATING MARGIN CONSTRAINT

Margin = (TCV - Adjusted COSP - Allocated SG&A) / TCV

MARGIN ≥ 16.40%

Hard constraint before quote emission; commercial policy defines the operative floor.

MODEL GOVERNANCE **Treat model outputs as decision support. Keep pricing rules, floor enforcement, approval routing, and audit logging deterministic.**

INDUSTRY LENS

Use one pricing platform, but preserve industry-specific value signals

Segment the feature store and commercial logic across the four operating groups identified in the blueprint.

Financial Services

RISK + CONTROL

High regulatory risk and legacy core-banking density. Value signal: confidence in risk-controlled delivery.

CMT

VELOCITY + CHURN

Fast-moving, commoditized markets. Value signal: milestone pricing linked to subscriber churn reduction.

Products (Retail / CPG)

SUPPLY CHAIN

Seasonality and supply-chain priorities. Constraint: sensitivity to fixed cash commitments.

Resources (Energy / Utilities)

CAPITAL + COMMODITIES

Multi-year capital transformations. Input: rate cards indexed to relevant commodity benchmarks.

Common platform services Shared event pipeline • common margin engine • reusable-asset catalog • auditable approval policy

COMMERCIAL GOVERNANCE

Three-tier routing turns pricing policy into an executable workflow

Discount and margin thresholds determine the approval path; exceptional deals cannot bypass the audit ledger.

TIER 1

Instant AI clearance

TRIGGER

Discount ≤ 8.0%
AND margin ≥ 16.5%
AND (reuse α ≥ 20% OR Option B)

DECISION SLA **< 1.8 seconds**

Issue clearance token; log to Spanner; send Salesforce webhook to generate order forms.

TIER 2

Regional commercial review

TRIGGER

Discount 8.1%-12.0%
OR margin 15.0%-16.4%

DECISION SLA **4-hour SLA**

Queue for regional lead. Cloud Tasks timer escalates to Tier 3 if the deadline expires.

TIER 3

Global CFO / committee

TRIGGER

Discount > 12.0%
OR margin < 15.0%

DECISION SLA **24-hour SLA (source)**

Run Monte Carlo sensitivity replay; await executive callback; require dual FIDO2 hardware-token sign-off.

POLICY HARDENING **Define one canonical threshold table and rounding convention for production rules before release.**

EXECUTION SEQUENCE

A quote moves from inference to clearance - or timed escalation

Illustrative source scenario: a $50M GFS deal with asset reuse estimate α = 34.2%.

1

Submit quote

Salesforce CPQ sends scope, terms, client and deal metadata.

2

Infer value

Vertex AI returns WTP curve and reuse estimate (α = 34.2%).

3

Evaluate rules

Cloud Run pricing engine computes margin and applies policy.

4

Route decision

Cloud Workflows selects Tier 1, Tier 2 or Tier 3.

5

Record outcome

Write token / approval / escalation evidence to the audit ledger.

DECISION BRANCH

TIER 1 **Eligible quote**

Token issued; clearance in under 2 seconds.

TIER 2 **Regional review**

4-hour timer; expired review escalates to Tier 3.

TIER 3 **Executive override**

Monte Carlo replay + dual hardware-token sign-off.

The α value is an input to the cost-reuse model; it does not by itself authorize the quote.

SECURITY + GOVERNANCE

Bake identity, data protection and auditability into the commercial path

Protect pricing history, rate cards and approval evidence with layered controls.

Perimeter

VPC-SC

VPC Service Controls perimeter around BigQuery, Vertex AI, Cloud Storage and Cloud Run to reduce exfiltration risk.

Identity

OIDC + IAM

Google-signed OIDC service-to-service bearer tokens; per-service identities and least-privilege permissions.

Encryption

CMEK

Customer-managed Cloud KMS encryption keys for pricing telemetry at rest.

Audit

IMMUTABLE LOGS

Commercial decisions and clearance tokens recorded to Cloud Spanner and immutable BigQuery audit logs.

Executive approval

FIDO2

Dual hardware-token authentication for Tier 3 sign-off and executive override.

Human control

HITL

Regional lead review for exceptions, timed escalation, and a clear evidentiary trail.

Control objective **Every quote decision should be attributable, explainable, policy-checked and recoverable from its audit record.**

FINOPS + OPERABILITY

Make cost elasticity measurable, and verify every service-level target

The architecture proposes a small interactive footprint and shifts expensive enrichment workloads away from peak quote paths.

RIGHT-SIZE RUNTIME

Cloud Run instances: 1 vCPU / 512 MiB; concurrency 80. Validate memory and CPU headroom under realistic Monte Carlo workloads.

MOVE HEAVY JOBS OFF-PEAK

Model retraining, embeddings and batch prediction run overnight; use BigQuery scheduled queries during lower-cost windows.

CONTROL QUERY COST

Partition BigQuery tables by deal_submission_date and cluster by operating_group and client_id.

VERIFICATION GATES

01 **Latency**

Prove p99 < 200 ms under load; source target states < 85 ms for warm services.

02 **Margin floor**

Reject or route quotes that fail the operative minimum-margin rule.

03 **SLA escalation**

Test that the 4-hour review timer escalates missed deadlines reliably.

04 **Audit completeness**

Reconcile quote, inference, approval, token and final contract records.

Cost target from source: under $0.002 per simulation. Benchmark against actual query shapes, data volume and workload concurrency before adopting as an SLO.

IMPLEMENTATION GATES

Lock the policy and prove the targets before global rollout

The supplied blueprint sets a strong direction; four areas need explicit owner sign-off to make it production-ready.

01	**Commercial policy**	Publish one canonical margin-floor value and one threshold table for discount, margin, asset reuse and Option B eligibility.
02	**Boundary behavior**	Specify percentage precision, rounding, missing-data behavior and precedence when discount and margin thresholds point to different tiers.
03	**Performance evidence**	Load-test the proposed 1 vCPU / 512 MiB container profile, concurrency 80, warm pool and multi-region behavior; confirm p99 and failover.
04	**Model governance**	Define monitoring, drift review, evaluation datasets, explainability expectations, fallback behavior and human override procedures.

READY-TO-RELEASE CHECKLIST

- Thresholds signed off by Commercial + CFO
- Load and failover tests pass
- Security perimeter and IAM reviewed
- Audit replay and escalation tested
- AI models monitored with safe fallback

Source: supplied Enterprise Architecture Blueprint. Financial, performance and scale figures should be treated as stated design assumptions until validated.